

# **IoT-Based Smart Dengue Environmental Risk Monitoring and Early Warning System Using ESP32**

A low-cost IoT approach for real-time environmental risk monitoring

## **Project Proposal**

**Track A** (Embedded IoT System)

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GitHub repository: <https://github.com/Rushdhy18/iot-smart-dengue-risk-monitoring.git>

# 1. Introduction / Background

Dengue is influenced by environmental and climatic conditions such as **temperature, rainfall and humidity**, while stagnant water provides potential breeding environments for *Aedes* mosquitoes. Research on dengue early-warning systems has therefore increasingly incorporated meteorological, environmental and epidemiological information.

However, many existing approaches depend on **historical datasets, weather stations, epidemiological records, or centralized data sources** rather than directly measuring environmental conditions at a specific location.

Our system uses an **ESP32-based IoT device** to continuously monitor:

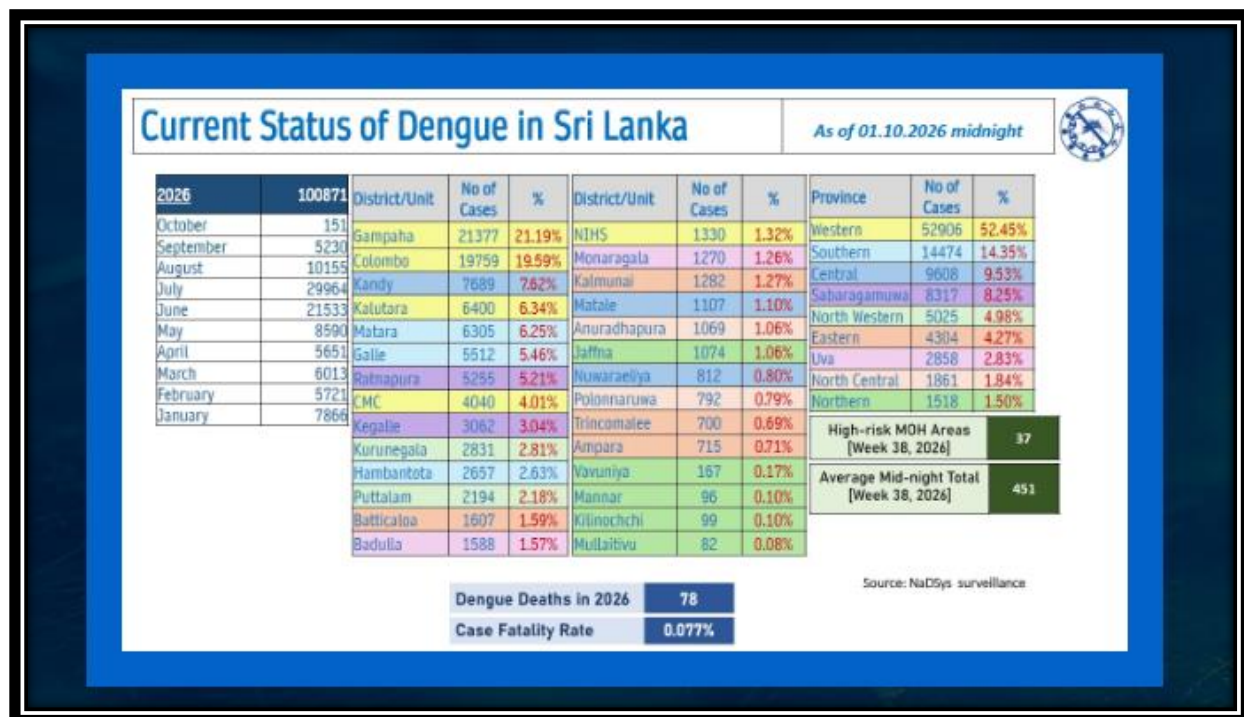
- *Temperature*
- *Humidity*
- *Rainfall*
- *Stagnant-water depth*
- *Historical risk index*

These measurements are converted into a **0–100 environmental risk score** and transmitted through MQTT for future visualization and analysis.

## 2. Problem Statement(5W1H)

| Dimension | Our project                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WHO       | Primary: Public Health Inspectors (PHIs) and Medical Officers of Health (MOH) responsible for vector control in the <b>Gampaha and Colombo districts</b> . These two districts recorded 38,945 of Sri Lanka's 94,666 dengue cases in 2026 up to 26 August, about 41% of the national total (Gampaha 20,170, Colombo 18,775) [1].<br>Secondary: residents of the monitored zone and local-authority clean-up teams, including the Colombo Municipal Council.                                             |
| WHAT      | There is no continuous, localized measurement of dengue-favorable conditions (warm, humid weather and stagnant water after rain). Risk is inferred from periodic surveys and centralized case data. The burden is large and rising in 2017 Gampaha's incidence was 1,323 per 100,000 against a 2012-2016 mean of 252 (5.2 times higher), and Colombo's was 1,419 against 503 (2.8 times higher) [2]. In 2026 up to 26 August, Gampaha and Colombo were the two worst-affected districts nationally [1]. |
| WHERE     | Dense residential lanes with open drains and water-holding containers (buckets, tires, drums, gutters, blocked canals) in urban and peri-urban areas of Gampaha and Colombo districts. The setting is outdoor, humid and rain-exposed, in the wet zone with annual rainfall above 2,500 mm [2]. Wi-Fi is available at or near the node.                                                                                                                                                                 |
| WHEN      | Continuously, with conditions changing within hours to days of rainfall. Cases follow the rains: 2026 cases peaked in July at 29,962 [1], and the NDCU linked the surge to increased rainfall and flooding in the southwest monsoon season [5]. A node raises an alert when water persists: <b>depth of at least 2 cm after at least 20</b>                                                                                                                                                             |

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <b>mm of rain in 24 h, still present at 5 days (120 h) for a warning and at 7 days (168 h) for a critical alert.</b> The <i>Aedes</i> cycle from egg to adult takes about 7-10 days, so the 5-day warning comes before the earliest adult emergence [3].                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>WHY</b> | Late detection delays inspection and larval control. Entomological surveillance relies on routine cluster surveys done monthly [2], while the breeding cycle lasts only 7-10 days [3], so several cycles can finish between visits. Mosquitoes need only a small amount of water to lay eggs [3]. Enforcement also depends on inspecting premises, since MOH offices can take legal action against premises that neglect breeding sites [5]. These methods are labor-intensive and cannot see local changes between visits. Burden: Sri Lanka recorded 49,887 dengue cases and 24 deaths in 2024 [4], and 94,666 cases and 72 deaths in 2026 up to 26 August, with 76 high-risk MOH areas in week 33 [1]. |
| <b>HOW</b> | No change: DHT22, rainfall and ultrasonic sensing, then edge risk scoring on the ESP32, local alerts (LCD, LEDs, buzzer), and MQTT to a broker and dashboard, with a stagnation clock that warns at day 5 and day 7.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



The main problem is that dengue risk monitoring can depend on **periodic surveys and existing surveillance data**, which may not provide continuous, localized environmental information.

Our project addresses this by asking:

**Can a low-cost IoT device continuously collect localized environmental parameters and convert them into an interpretable dengue environmental risk indicator for early warning?**

### 3. Research Gap

Existing research shows that temperature, rainfall and humidity are frequently used in dengue prediction models. Research on dengue early-warning systems also shows that combining climatic, environmental and epidemiological information can improve surveillance, but **data quality, availability and integration remain important challenges**.

A recent South Asian review found **few robust, evidence-based operational forecasting systems**, and specifically identified opportunities for contextually relevant climate-driven early-warning tools in countries including Sri Lanka.

**“Although existing studies have developed dengue prediction and early-warning models using climatic and epidemiological datasets, there is a gap in low-cost, localized IoT systems that directly measure environmental conditions, particularly stagnant-water conditions, and provide real-time risk information.”**

### 4. Aim

- To develop a low-cost IoT-based system for real-time monitoring of environmental conditions associated with dengue risk and generation of an early-warning risk indicator.

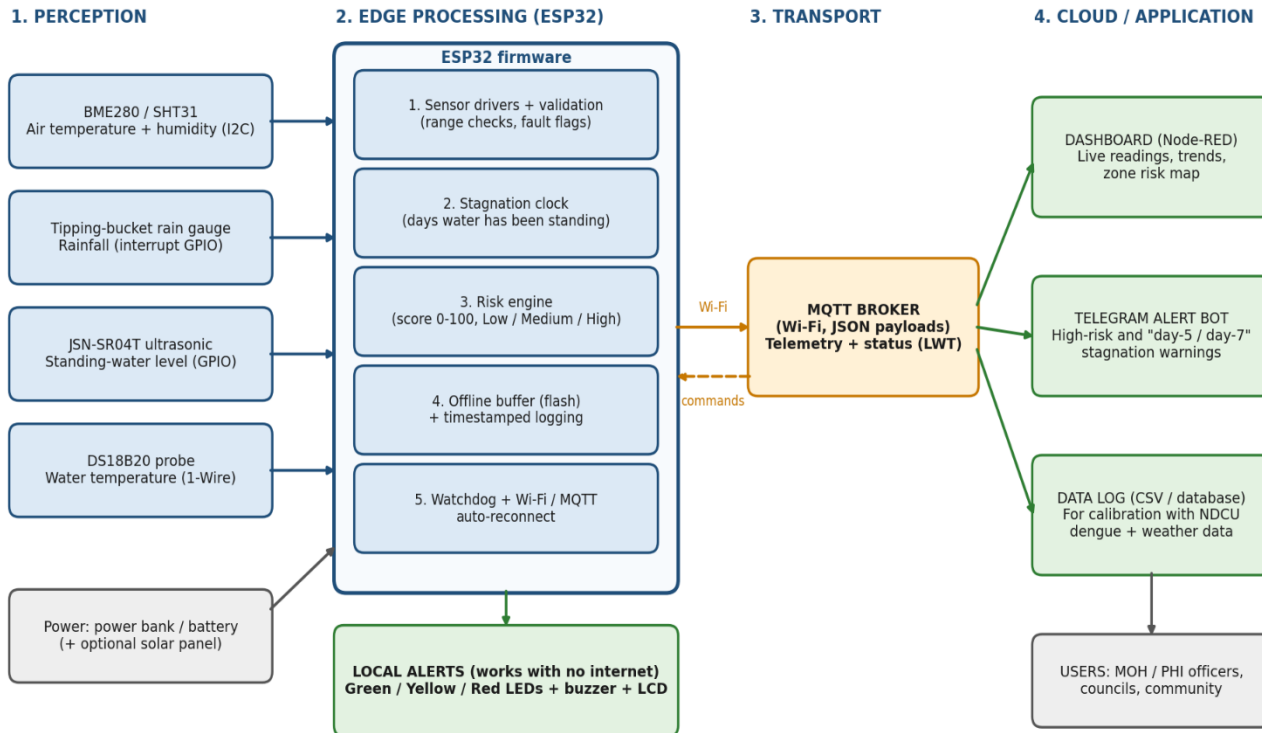
### 5. Objectives

1. Develop an ESP32-based environmental monitoring prototype.
2. Measure temperature and humidity using a DHT22 sensor.
3. Simulate rainfall and measure stagnant-water depth.
4. Develop a weighted environmental dengue-risk scoring model.
5. Transmit real-time measurements and risk information through MQTT for future GIS/dashboard integration.

### 6. Target Users and Use Cases

- **Public Health Inspector (primary):** checks risk levels and zone coordinates from the CMC area nodes to prioritize which school compounds, drains and premises to inspect first.
- **School staff or residents:** see the local risk level on the node (LCD, LEDs, buzzer) when rain and stagnant water raise the score, and clear the water.
- **Monitoring authority / researcher:** consumes MQTT data in a database and GIS dashboard for risk visualization by location (future work).

## 7. System Design



### 7.1 Layers and components

| Layer           | Components                                                                                                                               | Responsibility                                                                                       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Sensing         | BME280 or SHT31 (air temperature, humidity), tipping-bucket rain gauge, JSN-SR04T waterproof ultrasonic sensor, DS18B20 waterproof probe | Measure the four breeding drivers: air conditions, rainfall, standing-water level, water temperature |
| Edge processing | ESP32 firmware (five modules, Figure 1)                                                                                                  | Validate readings, track stagnation time, compute risk, buffer data, recover from faults             |
| Local output    | Green / yellow / red LEDs, buzzer, 16x2 LCD                                                                                              | Immediate warning at the site even with no internet                                                  |
| Communication   | Wi-Fi, MQTT broker, JSON payloads                                                                                                        | Send telemetry and status, receive commands                                                          |
| Presentation    | Node-RED dashboard, Telegram bot, CSV/database log                                                                                       | Live view, remote alerts, data for calibration                                                       |
| Power           | Power bank or battery, optional solar panel                                                                                              | Field operation                                                                                      |

## 7.2 Data Flow

- **Sample.** Every 5 seconds the ESP32 samples air temperature, humidity and water level. The rain gauge works by interrupt: each bucket tip adds a fixed amount of rainfall. Water temperature is read every minute.
- **Validate.** Each reading is checked against a valid range. Failed or missing readings are flagged and excluded from the score.
- **Process.** The stagnation clock updates, then the risk engine computes a score from 0 to 100.
- **Local alert.** LEDs and buzzer update immediately.
- **Publish or buffer.** A JSON message is published to the MQTT broker. If the link is down, it is stored in flash with a timestamp and uploaded when the link returns.
- **Present.** The dashboard shows the data, the Telegram bot sends High-risk and stagnation warnings, and every message is logged for later calibration.

## 8. Key logic

### Stagnation clock (the core innovation)

Aedes eggs develop into adult mosquitoes in roughly a week, which is why dengue campaigns advise emptying containers weekly. Instead of only reporting that water is present, the node starts counting when a rain event fills the reference container, and it warns before the week is over:

- Rain of at least a set amount within 24 hours and water depth above a set threshold starts the counter.
- Day 5 gives an amber warning ("empty or treat standing water").
- Day 7 gives a red warning.
- If the water level drops back below the threshold, the counter resets.

## 9. Risk engine

- The score combines normalised scores for air temperature (about 25-32 C is best for breeding), humidity, recent rainfall, standing-water depth and stagnation days, with a historical dengue index for the area. Levels are Low (below 40), Medium (40-69) and High (70 and above). The weights start as an explainable heuristic and will be calibrated with Gampaha / Colombo weather and dengue data.

| Parameter        | Weight |
|------------------|--------|
| Temperature      | 25%    |
| Humidity         | 20%    |
| Rainfall         | 20%    |
| Stagnant water   | 25%    |
| Historical index | 10%    |

$$\text{Risk} = 100(0.25T_s + 0.20H_s + 0.20R_s + 0.25W_s + 0.10H_i)$$

Then:

- 0 – <40      LOW
- 40 – <70     MEDIUM
- 70 – 100     HIGH

### 9.1MQTT Design

| Topic                      | Purpose                                                                                            |
|----------------------------|----------------------------------------------------------------------------------------------------|
| dengue/<node-id>/telemetry | Readings, risk score, level, stagnation days (JSON)                                                |
| dengue/<node-id>/status    | Online / offline. The broker publishes "offline" automatically (Last Will) if the node disconnects |
| dengue/<node-id>/cmd       | Optional commands to the node, such as sampling interval or thresholds                             |

### 9.2Draft pinout (ESP32 DevKit)

| Device                       | ESP32 pin                  | Note                                                        |
|------------------------------|----------------------------|-------------------------------------------------------------|
| BME280 / SHT31 and LCD (I2C) | GPIO21 (SDA), GPIO22 (SCL) | Shared bus; different I2C addresses                         |
| Rain gauge (reed switch)     | GPIO4                      | Internal pull-up, falling-edge interrupt, software debounce |
| JSN-SR04T TRIG / ECHO        | GPIO5 / GPIO18             | ECHO is 5 V, so use a voltage divider to 3.3 V              |
| DS18B20 data                 | GPIO15                     | 4.7 k pull-up to 3.3 V                                      |
| Green / yellow / red LEDs    | GPIO25 / 26 / 27           | 220 ohm series resistors                                    |
| Buzzer                       | GPIO14                     | Active buzzer through the pin or a transistor               |

## 9.3 Electrical integration (hardware build)

- Power: 5 V from power bank into the ESP32 board regulator (3.3 V LDO); 100 nF + 10 uF decoupling near sensors.
- Logic levels: HC-SR04 / JSN-SR04T ECHO is 5 V, so a 1 k / 2 k divider brings it to 3.3 V for the ESP32.
- I2C: BME280 and LCD share GPIO21 / GPIO22 on different addresses (0x76 and 0x27), using module pull-ups.
- 1-Wire: DS18B20 on GPIO15 with a 4.7 k pull-up to 3.3 V.
- Rain gauge: reed switch on GPIO4 with internal pull-up, falling-edge interrupt and 50 ms software debounce.
- LEDs: 220 ohm series resistors; buzzer driven directly or through a transistor.

## 10. Technology stack

| Category            | Items                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| Microcontroller     | ESP32 DevKit                                                                                                               |
| Sensors             | DHT22 (BME280 / SHT31 planned), potentiometer (tipping-bucket gauge planned), HC-SR04 (JSN-SR04T planned), DS18B20 planned |
| Outputs             | Green / yellow / red LEDs, buzzer, 16x2 I2C LCD                                                                            |
| Firmware            | Arduino framework (C++), libraries: PubSubClient, DHT, LiquidCrystal I2C; Wokwi simulator                                  |
| Communication       | Wi-Fi, MQTT (HiveMQ public broker for the prototype)                                                                       |
| Cloud / application | MQTT client dashboard now; Node-RED and Telegram bot planned; CSV log for calibration                                      |
| Version control     | GitHub: /firmware, /hardware, /docs, /tests                                                                                |

## 11. Known limitations

- One node measures one location; the standing-water measurement represents one reference container, not every site in an area.
- Risk weights are a heuristic until calibrated against real case data, and area-level case data may be limited.
- Public MQTT brokers are not secure; a private broker or authentication will be used for the final demo.

## 12. Expected Contribution

### 1. Localized monitoring

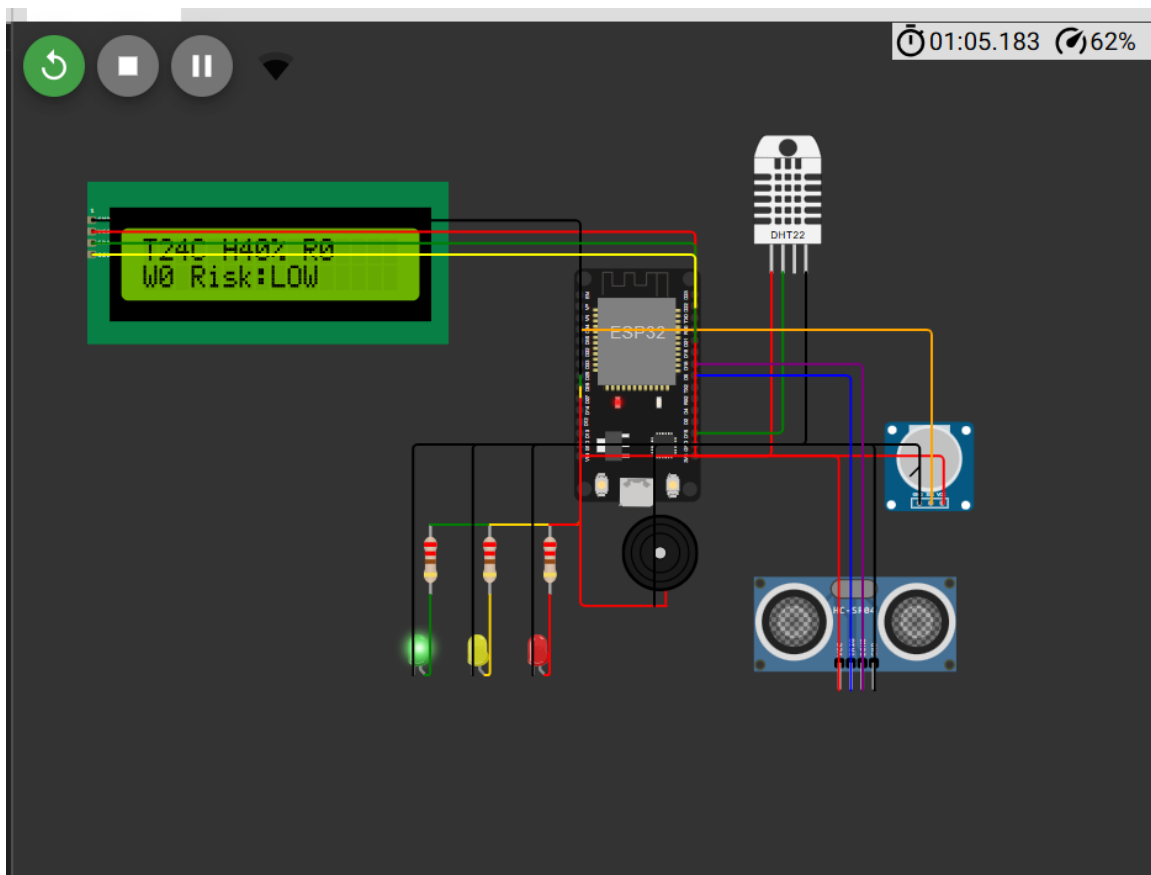
The system measures environmental conditions directly at the monitoring location.

### 2. Low-cost IoT architecture

ESP32 + sensors + MQTT provides a relatively simple platform for environmental monitoring.

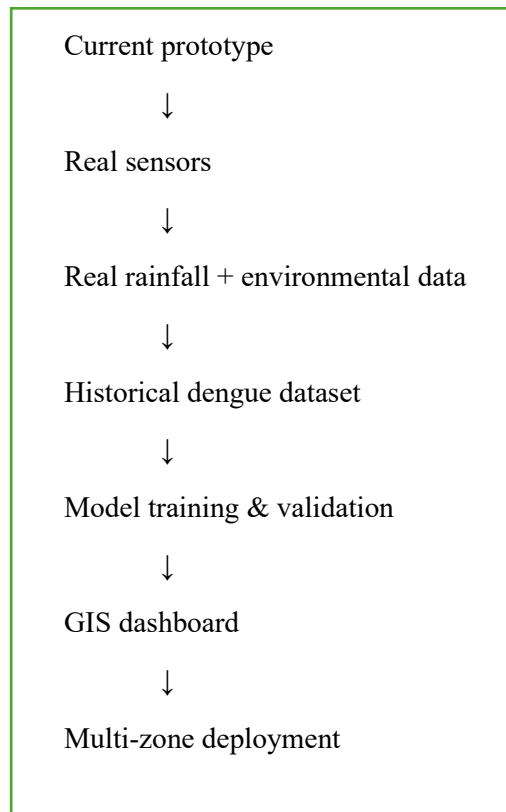
## 11. Testing

This is our systematic diagram simulated in wokwi





## 14. Future Work



- Future ML models could include approaches such as Random Forest or LSTM, which are among methods studied in dengue forecasting research.

## References

1. NDCU NaDSys data via Counterpoint, "Gampaha tops Lanka's dengue cases", 27 Aug 2026. <https://counterpoint.lk/gampaha-tops-lanka-s-dengue-cases/>
2. "Severe Dengue Epidemic, Sri Lanka, 2017", *Emerging Infectious Diseases* 2020;26(4):682-691, Table 1 and surveillance methods. <https://wwwnc.cdc.gov/eid/article/26/4/19-0435>
3. Epidemiology Unit, Sri Lanka, *Epidemiological Bulletin* vol. 45 no. 4-5 (2017 dengue epidemic). [https://www.epid.gov.lk/storage/post/pdfs/vol\\_45\\_no\\_05-english.pdf](https://www.epid.gov.lk/storage/post/pdfs/vol_45_no_05-english.pdf)
4. CDC, "Life Cycle of Aedes Mosquitoes". <https://www.cdc.gov/mosquitoes/about/life-cycles/aedes.html>
5. Ada Derana, 15 Jan 2025, quoting the NDCU. <https://adaderana.lk/news/104971/over-2000-dengue-cases-reported-in-sri-lanka-this-year->
6. Newsfirst, "Dengue cases surge past 90,000", 11 Aug 2026. <https://newsfirst.lk/2026/08/11/dengue-cases-surge-past-90-000-in-sri-lank>